

The three board problems

Rotating rings, two-frame joints, and terminal rewards

Working note

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Abstract

Three problems were circled on the board: a rotating point near a ring whose whole trajectory is one data point, the two-frame joint law and its Gaussian closed form, and a reward on the last frame propagated backwards. This note develops all three in the simplest setting that keeps their content, shows every computation, and states plainly which claims are proved, which are measured, and which turned out to be false. The central technical result is that the rotating ring model — previously recorded as having no closed-form joint score — is exactly solvable: the rotation is a gauge, the ring is a location mixture, and the joint score is a linear part plus a rank-one correction obtained from one one-dimensional Bessel integral. Two scaling claims made in a first pass are rejected here by measurement.

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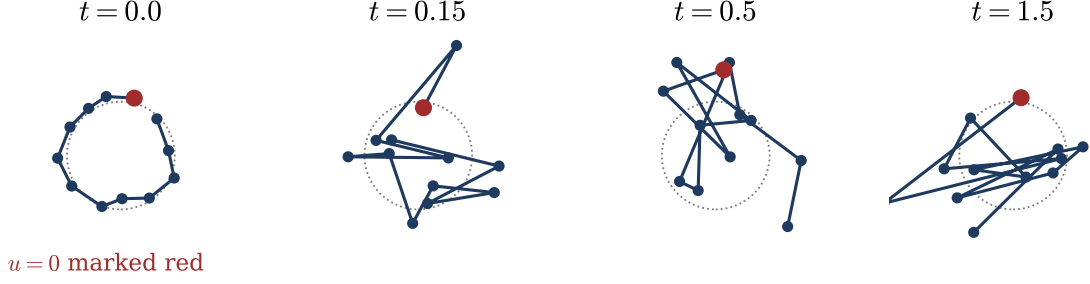


Figure 1: The two clocks. One trajectory of the rotating ring model ($T = 12$, $\sigma = 0.10$, $\psi = 2\pi/12$), shown at four diffusion times t . The internal time u runs along the polyline, from the red dot. The diffusion time t shrinks the whole trajectory towards the origin and blurs it; at $t = 1.5$ nothing of the dynamics survives. The dotted circle is the unit ring.

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1 The setting, and why the joint

1.1 Two clocks

Everything in this note has two time variables and keeping them apart is the whole discipline of the problem.

- The *internal* time $u = 0, 1, \dots, T - 1$ indexes position along a trajectory. On the board it appears as $u \in [0, 2\pi/\omega]$ with the working value $T = 40$.
- The *diffusion* time t indexes the noise level of the generative channel.

A data point is not a frame. It is a *whole trajectory*,

$$a = (x(0), y(0), x(1), y(1), \dots, x(T-1), y(T-1)) \in \mathbb{R}^{2T}, \quad D = 2T, \quad (1)$$

exactly as the board writes it, $a^\mu = \{x^\mu(u), y^\mu(u)\}$. The forward channel is the variance-preserving Ornstein–Uhlenbeck channel,

$$x = m a + \sqrt{\Delta} \xi, \quad m = e^{-t}, \quad \Delta = 1 - e^{-2t}, \quad \xi \sim \mathcal{N}(0, I_D), \quad (2)$$

and it acts on *all* of a at once. That single fact is why the object of study is the *joint* score

$$s(x, t) = \nabla_x \log P_t(x), \quad P_t(x) = \int da P_0(a) \mathcal{N}(x; ma, \Delta I_D), \quad (3)$$

the board's $s(x, t) \rightarrow \nabla \log(P_0 \otimes G)$, and not a collection of per-frame scores. Figure 1 shows the two clocks acting on one trajectory: the internal time runs along the curve, the diffusion time destroys it.

1.2 The score–posterior identity

We use one standard identity throughout. Differentiating (3),

$$\nabla_x P_t(x) = \int da P_0(a) \mathcal{N}(x; ma, \Delta I) \frac{ma - x}{\Delta}, \quad (4)$$

so dividing by $P_t(x)$,

$$\boxed{s(x, t) = \frac{m \mathbb{E}[a \mid x] - x}{\Delta}} \quad (5)$$

where the expectation is over the posterior of the clean trajectory given the noisy one. The score is a denoiser. Every closed form below is, at bottom, a posterior mean that happens to be computable.

2 Problem 1: the rotating ring

2.1 The model

The board’s left panel. Three latent variables, independent:

$$r_0 \sim p_\lambda(\rho) \propto \rho e^{-(\rho-1)^2/2\lambda}, \quad \theta_0 \sim \text{Unif}[0, 2\pi), \quad \psi \sim p(\psi), \quad (6)$$

and then

$$z_0 = r_0(\cos \theta_0, \sin \theta_0), \quad z_{u+1} = R_\psi z_u + \sigma \eta_{u+1}, \quad \eta_u \sim \mathcal{N}(0, I_2) \text{ i.i.d.} \quad (7)$$

Here R_ψ is planar rotation by $\psi = \omega \Delta u$. The factor ρ in p_λ is the two-dimensional area element: the density of the point $z_0 \in \mathbb{R}^2$ is $\propto e^{-(\|z\|-1)^2/2\lambda}$, so its radial marginal carries an extra ρ . Figure 2 checks the sampler against this.

(7) is the discrete form of the pair written on the board, $dr = (1-r)du + \Gamma dB_u$ from the potential $V(r) = \frac{1}{2}(r-1)^2$, together with $\dot{\theta} = \omega$.

At $T = 2$ the data vector is the board’s own $(x_0, y_0, x_1, y_1) = a^\mu \in \mathbb{R}^4$. With three latents in ambient dimension four, this is the board’s “ ρ dim 3”. That is the setting we use.

2.2 The symmetry

P_0 is invariant under rotating *every* frame by the same angle, $\mathcal{R}_\phi = \text{blockdiag}(R_\phi, \dots, R_\phi)$. Indeed $\mathcal{R}_\phi a$ is the trajectory built from $(r_0, \theta_0 + \phi, \psi)$, and θ_0 is uniform. Since the channel (2) is isotropic, P_t inherits the invariance for every t , and the score is equivariant,

$$s(\mathcal{R}_\phi x, t) = \mathcal{R}_\phi s(x, t). \quad (8)$$

This is the board’s boxed $R_\psi(x)$ statement. It has a strong consequence, Theorem 2 below.

2.3 Step 1: the rotation is a gauge

Lemma 1 (Gauge reduction). *Let $U_\psi = \text{blockdiag}(R_{-\omega\psi})_{u=0}^{T-1} \in O(2T)$. Then $U_\psi a$ has the law of the same model with $\psi = 0$, i.e. a planar random walk started on the ring. Consequently, writing P_t^0 and s^0 for the $\psi = 0$ quantities,*

$$P_t(x \mid \psi) = P_t^0(U_\psi x), \quad s(x, t \mid \psi) = U_\psi^\top s^0(U_\psi x, t). \quad (9)$$

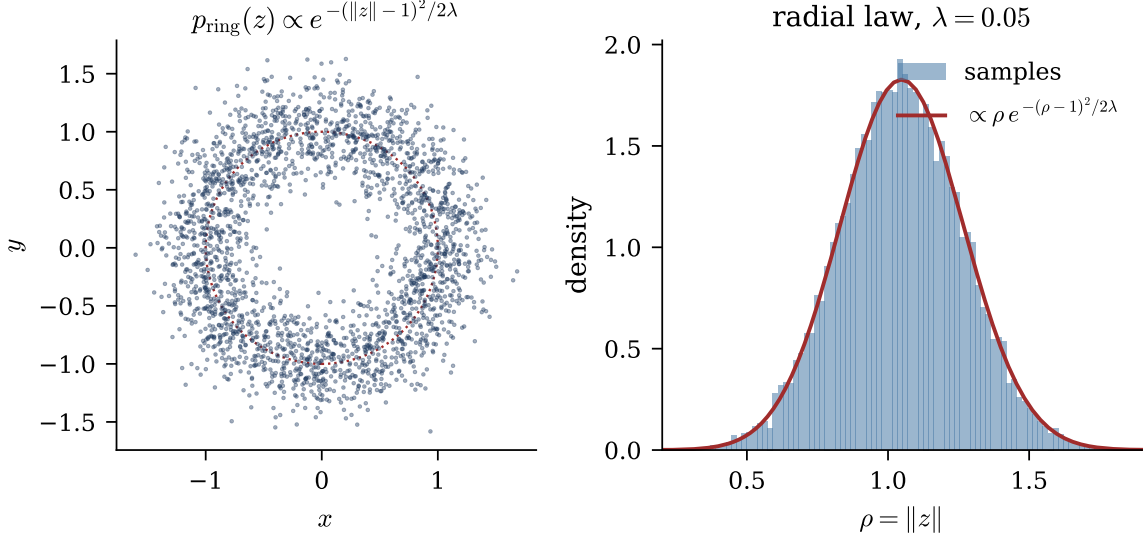


Figure 2: The ring prior, $\lambda = 0.05$. Left: 2500 draws of z_0 with the unit circle dotted. Right: the histogram of $\rho = \|z_0\|$ from 20 000 draws against the radial density $\propto \rho e^{-(\rho-1)^2/2\lambda}$. The mode sits slightly outside $\rho = 1$ because of the area element.

Proof. Put $\tilde{z}_u = R_{-u\psi} z_u$. Then from (7)

$$\tilde{z}_{u+1} = R_{-(u+1)\psi} (R_{\psi} z_u + \sigma \eta_{u+1}) = R_{-u\psi} z_u + \sigma R_{-(u+1)\psi} \eta_{u+1} = \tilde{z}_u + \sigma \tilde{\eta}_{u+1}, \quad (10)$$

with $\tilde{\eta}_{u+1} = R_{-(u+1)\psi} \eta_{u+1}$. A rotation of an isotropic Gaussian is the same isotropic Gaussian, and the $\tilde{\eta}_u$ remain independent, so $\tilde{z}$ is a random walk with step σ ; and $\tilde{z}_0 = z_0$ still sits on the ring.

For the second claim, U_ψ is orthogonal, so applying it to (2) gives $U_\psi x = m U_\psi a + \sqrt{\Delta} U_\psi \xi$ with $U_\psi \xi \sim \mathcal{N}(0, I)$. Hence $U_\psi x$ has exactly the law of the noisy $\psi = 0$ model, and since $|\det U_\psi| = 1$ the densities satisfy $P_t(x | \psi) = P_t^0(U_\psi x)$. Differentiating by the chain rule gives the score identity. $\square$

Remark 1. At known ψ the rotation contributes *zero* statistical difficulty: it is a known orthogonal change of frame. Everything hard about the model is a random walk anchored on a ring. Figure 3 shows the two frames.

2.4 Step 2: the ring is a location mixture

Work in the gauged frame and write the trajectory as a matrix $X \in \mathbb{R}^{T \times 2}$, row u being frame u . From $\tilde{z}_u = z_0 + \sigma \sum_{k \leq u} \eta_k$ we get, *conditionally on* z_0 , a Gaussian with

$$\mathbb{E}[\tilde{z}_u | z_0] = z_0, \quad \text{Cov}(\tilde{z}_u, \tilde{z}_v | z_0) = \sigma^2 \min(u, v) I_2. \quad (11)$$

So with $K_{uv} = \min(u, v)$ the conditional law of the trajectory is $\mathcal{N}(\mathbf{1} z_0^\top, \sigma^2 K \otimes I_2)$, and after the channel

$$X | z_0 \sim \mathcal{N}(m \mathbf{1} z_0^\top, A_t \otimes I_2), \quad \boxed{A_t = m^2 \sigma^2 K + \Delta I_T} \quad (12)$$

The point is that A_t *does not depend on* z_0 . So P_0 , and P_t , is a two-parameter *location* mixture of a single Gaussian — the easiest kind of mixture there is.

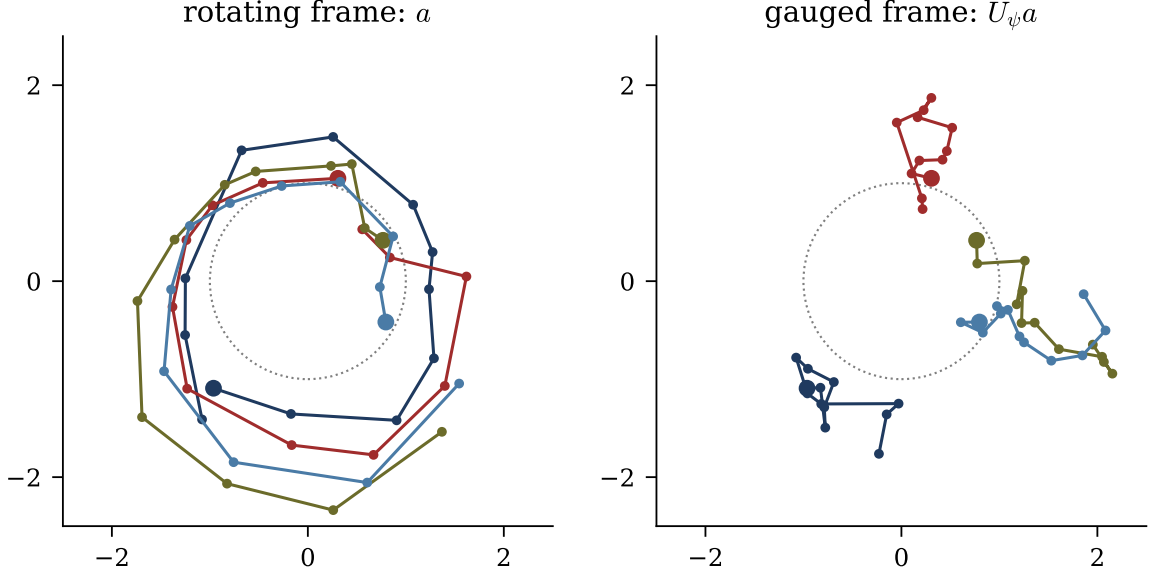


Figure 3: The gauge, on four trajectories ($T = 12$, $\sigma = 0.22$, $\psi = 30^\circ$; large dots mark $u = 0$). Left: the data, coherently circulating. Right: the same trajectories after U_ψ . The circulation is gone and what remains is a random walk started on the ring. U_ψ is orthogonal, so it commutes with the isotropic channel and this reduction is exact at every diffusion time.

2.5 Step 3: the exact joint score

Theorem 1 (Exact joint score, any T). *Let $g = A_t^{-1}\mathbf{1}$, $\kappa_t = \mathbf{1}^\top g$, $b = X^\top g \in \mathbb{R}^2$, $\beta = \|b\|$ and $\hat{b} = b/\beta$. Then*

$$\log P_t(X) = c(t) - \frac{1}{2} \text{tr}[X^\top A_t^{-1} X] + \log \mathcal{Z}_t(\beta), \quad (13)$$

$$\mathcal{Z}_t(\beta) = \int_0^\infty \rho e^{-(\rho-1)^2/2\lambda} e^{-m^2 \kappa_t \rho^2/2} I_0(m\rho\beta) d\rho, \quad (14)$$

and therefore

$$s(X, t) = -A_t^{-1} X + \Phi_t(\beta) g \hat{b}^\top, \quad \Phi_t = \partial_\beta \log \mathcal{Z}_t \quad (15)$$

a linear part plus a rank-one correction.

Proof. Start from the mixture and expand the quadratic form in the exponent. Using $\text{tr}[M^\top N]$ for the Frobenius pairing,

$$\begin{aligned} \text{tr}[(X - m\mathbf{1}z_0^\top)^\top A_t^{-1}(X - m\mathbf{1}z_0^\top)] &= \text{tr}[X^\top A_t^{-1} X] - 2m \text{tr}[z_0 \mathbf{1}^\top A_t^{-1} X] + m^2 \text{tr}[z_0 \mathbf{1}^\top A_t^{-1} \mathbf{1} z_0^\top] \\ &= \text{tr}[X^\top A_t^{-1} X] - 2m b \cdot z_0 + m^2 \kappa_t \|z_0\|^2, \end{aligned} \quad (16)$$

because $\text{tr}[z_0 \mathbf{1}^\top A_t^{-1} X] = \mathbf{1}^\top A_t^{-1} X z_0 = b \cdot z_0$ and $\text{tr}[z_0 \mathbf{1}^\top A_t^{-1} \mathbf{1} z_0^\top] = \kappa_t \|z_0\|^2$. Hence

$$P_t(X) = c(t) e^{-\frac{1}{2} \text{tr}[X^\top A_t^{-1} X]} \int_{\mathbb{R}^2} dz_0 e^{-(\|z_0\|-1)^2/2\lambda} e^{m b \cdot z_0 - \frac{1}{2} m^2 \kappa_t \|z_0\|^2}. \quad (17)$$

Now the integral is two-dimensional and its integrand depends on the direction of z_0 only through $b \cdot z_0$. Put $z_0 = \rho(\cos \phi, \sin \phi)$ so that $dz_0 = \rho d\rho d\phi$ and $b \cdot z_0 = \rho\beta \cos(\phi - \arg b)$. The angular integral is the standard integral representation of the modified Bessel function,

$$\int_0^{2\pi} e^{m\rho\beta \cos(\phi - \arg b)} d\phi = 2\pi I_0(m\rho\beta), \quad (18)$$

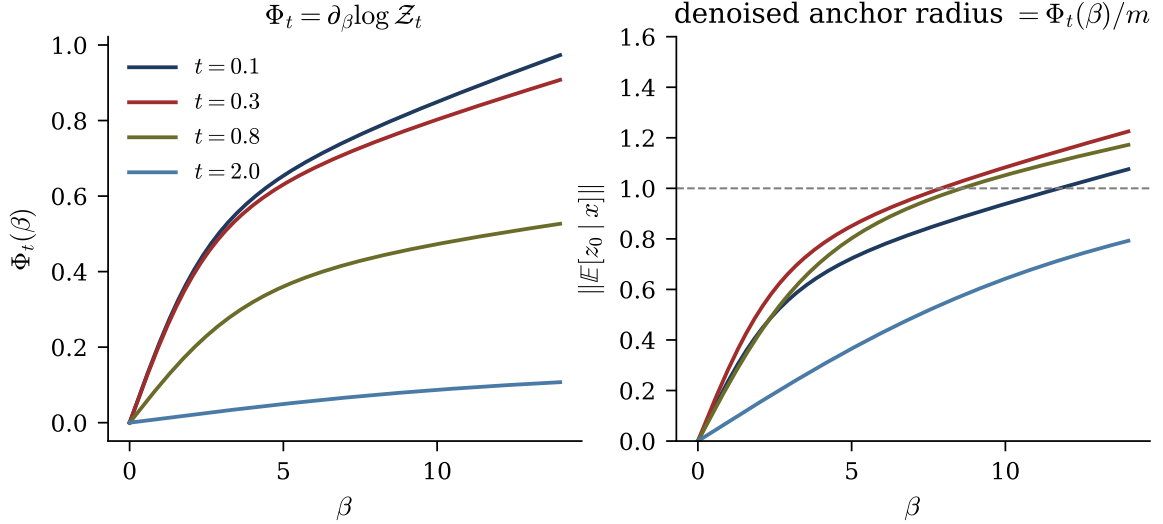


Figure 4: The scalar function that carries all the non-Gaussianity ($T = 6$, $\sigma = 0.30$, $\lambda = 0.05$). Left: $\Phi_t(\beta) = \partial_\beta \log \mathcal{Z}_t$. Right: the same divided by m , which by Proposition 1 is the length of the posterior mean of the ring anchor. It saturates at 1 (dashed) — once the data pin the anchor, it sits on the unit ring.

which leaves exactly $2\pi \mathcal{Z}_t(\beta)$ of (14) and proves (13).

For the gradient, the first term of (13) gives $-A_t^{-1}X$. For the second, $b_j = \sum_u g_u X_{uj}$, so $\partial b_j / \partial X_{ui} = \delta_{ij} g_u$ and

$$\frac{\partial \beta}{\partial X_{ui}} = \frac{b_j}{\beta} \frac{\partial b_j}{\partial X_{ui}} = g_u \hat{b}_i. \quad (19)$$

Therefore $\nabla_X \log \mathcal{Z}_t(\beta) = \Phi_t(\beta) g \hat{b}^\top$, an outer product, hence rank one. $\square$

So for *any* number of frames the exact joint score costs one one-dimensional integral. Two readings of the correction term:

Proposition 1 (The correction is the denoised anchor). $\Phi_t(\beta) \hat{b} = m \mathbb{E}[z_0 | X, t]$.

Proof. By (17) the posterior of z_0 given X is $\propto e^{-(\|z_0\|-1)^2/2\lambda} e^{mb \cdot z_0 - \frac{1}{2}m^2 \kappa_t \|z_0\|^2}$. Its normalisation is $2\pi \mathcal{Z}_t(\beta)$, which depends on b only through $\beta = \|b\|$. Differentiating the log-normalisation with respect to b pulls out exactly m times the posterior mean: $\nabla_b \log \mathcal{Z}_t(\|b\|) = m \mathbb{E}[z_0 | X]$. But $\nabla_b = \hat{b} \partial_\beta$, so $\Phi_t(\beta) \hat{b} = m \mathbb{E}[z_0 | X]$. $\square$

That is why the correction is rank one and why it points along $\hat{b}$: it is the single denoised ring anchor, broadcast to every frame with weights g . Figure 4 plots Φ_t and the anchor radius Φ_t/m : at small t the anchor sits on the ring, $\|\mathbb{E}[z_0 | x]\| \rightarrow 1$; at large t the posterior collapses to the origin.

2.6 The band structure of A_t^{-1}

A_t is where the temporal structure lives, so its inverse is worth looking at.

Proposition 2. $K_{uv} = \min(u, v)$ has rank $T - 1$ and vanishing 0th row, so

- (a) A_t is exactly block diagonal, $[\Delta] \oplus (\text{rest})$: frame 0 decouples in A_t^{-1} at every t ;
- (b) on the remaining block the d th off-diagonal of A_t^{-1} is of order $(\Delta/m^2 \sigma^2)^d$, so A_t^{-1} becomes tridiagonal as $t \rightarrow 0$.

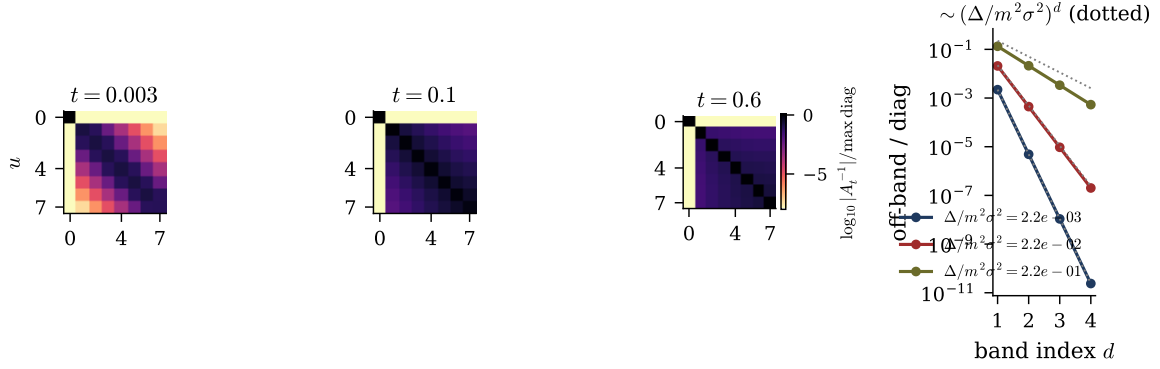


Figure 5: Band structure of A_t^{-1} ($T = 8$, $\sigma = 0.30$). First three panels: $\log_{10}$ of $|A_t^{-1}|$ normalised by its largest diagonal entry. Frame 0 is decoupled at every t (white cross), exactly. Right: the d th off-band, normalised, against the prediction $(\Delta/m^2\sigma^2)^d$ (dotted). The law holds over eleven decades.

Part (a) is not an accident: frame 0 *is* z_0 and carries no walk noise, so conditionally on z_0 it has zero variance and its only spread is the diffusion Δ . Note this also means A_0^{-1} does not exist; an earlier draft of this note asserted “ $A_0^{-1} = K^{-1}/\sigma^2$ is tridiagonal”, which is meaningless. Part (b) is the correct statement, and it is the band-fill law of the Gaussian chain (claim G8 of the main thesis ledger) in this model, with small parameter $\Delta/m^2\sigma^2$. Figure 5 measures it.

2.7 Step 4: a random rotation rate

The board wants ω drawn per sample. That costs one more one-dimensional integral and nothing else.

Proposition 3 (Random ψ). *For any T ,*

$$s(x, t) = \int d\psi p(\psi | x, t) U_\psi^\top s^0(U_\psi x, t), \quad p(\psi | x, t) \propto p(\psi) P_t^0(U_\psi x). \quad (20)$$

Proof. $P_t(x) = \int p(\psi) P_t(x | \psi) d\psi$, so

$$\nabla \log P_t = \frac{\int p(\psi) \nabla P_t(x | \psi) d\psi}{P_t(x)} = \int \frac{p(\psi) P_t(x | \psi)}{P_t(x)} \nabla \log P_t(x | \psi) d\psi, \quad (21)$$

which is (20) after inserting Lemma 1. $\square$

Combining Theorem 1 and Proposition 3: the full board model has **ground-truth score at any T from two nested one-dimensional quadratures** — the ψ integral outside, the Bessel integral inside. No network, no Monte Carlo, no curse of dimension. This is the tool everything below uses.

2.8 At two frames the posterior simplifies

Proposition 4. *At $T = 2$ only, $A_t = \text{diag}(\Delta_1, \Delta_2)$ with $\Delta_1 = \Delta$, $\Delta_2 = m^2\sigma^2 + \Delta$, and*

$$p(\psi | x, t) \propto p(\psi) \mathcal{Z}_t(\beta(\psi)), \quad \beta(\psi)^2 = \frac{|x_0|^2}{\Delta_1^2} + \frac{|x_1|^2}{\Delta_2^2} + \frac{2|x_0||x_1|\cos(\gamma - \psi)}{\Delta_1\Delta_2}, \quad (22)$$

where $\gamma = \arg x_1 - \arg x_0$ is the observed turning angle. The data enter only through $(|x_0|, |x_1|, \gamma)$, and the posterior is unimodal with its peak at $\psi = \gamma$.

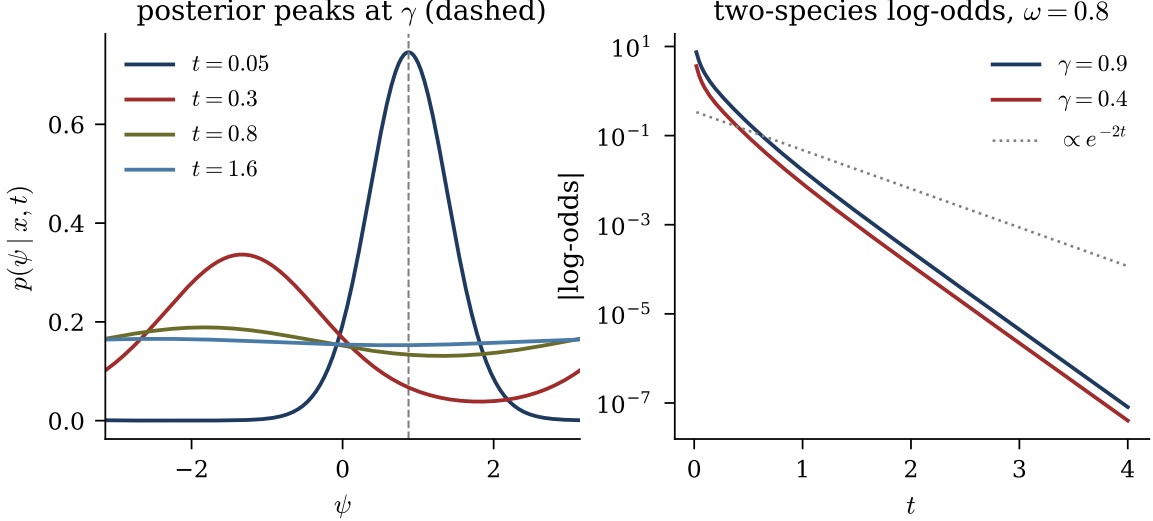


Figure 6: Left: the ψ -posterior at $T = 2$ for one data point, at four diffusion times. It peaks at the observed turning angle γ (dashed) and flattens as t grows: the dynamics are being forgotten. Right: the two-species log-odds of (25) for two values of the true turning angle, against the predicted e^{-2t} decay (dotted). A trajectory with $\gamma = 0.4$ is intrinsically harder to classify than one with $\gamma = 0.9$, as the $\sin \gamma$ factor requires.

Proof. For $T = 2$, $K = \text{diag}(0, 1)$, so A_t is diagonal as stated. Then

$$\text{tr}[(U_\psi X)^\top A_t^{-1} (U_\psi X)] = \sum_u \frac{\|R_{-u\psi} x_u\|^2}{(A_t)_{uu}} = \sum_u \frac{\|x_u\|^2}{(A_t)_{uu}}, \quad (23)$$

which is ψ -free because rotations preserve norms. So in (13) only $\log \mathcal{Z}_t(\beta(\psi))$ depends on ψ , giving the first claim. For the second, use complex coordinates: $b = x_0/\Delta_1 + e^{-i\psi} x_1/\Delta_2$, so

$$|b|^2 = \frac{|x_0|^2}{\Delta_1^2} + \frac{|x_1|^2}{\Delta_2^2} + \frac{2}{\Delta_1 \Delta_2} \text{Re}(\bar{x}_0 e^{-i\psi} x_1), \quad (24)$$

and $\text{Re}(\bar{x}_0 e^{-i\psi} x_1) = |x_0| |x_1| \cos(\gamma - \psi)$. Finally $\mathcal{Z}_t$ is a positive mixture of $I_0(m\rho\beta)$, each increasing in β , so $\mathcal{Z}_t$ increases in β , and $\beta(\psi)$ is maximised at $\psi = \gamma$. $\square$

Remark 2 (Scope). The simplification is genuinely special to $T = 2$. U_ψ conjugates $A_t \otimes I_2$ into itself if and only if A_t is diagonal in u , and $K_{uv} = \min(u, v)$ is diagonal only for $T = 2$. For $T \geq 3$ the cross terms $x_u \cdot R_{(u-v)\psi} x_v$ survive and the quadratic form does depend on ψ ; measured, the spread is 10–13% at $T = 3, 5$. The audit tests both sides of this boundary. Proposition 3 is unaffected — it is Bayes, and holds for all T .

Figure 6, left, is (22) in action: a bump on the observed turning angle that widens as the noise rises.

2.9 Speciation: committing to a sense of rotation

Take two species, clockwise and counter-clockwise, $p(\psi) = \frac{1}{2}(\delta_{+\omega} + \delta_{-\omega})$. By Proposition 4 the order parameter is one scalar,

$$\text{logit } \Pr(\psi = +\omega \mid x, t) = \log \mathcal{Z}_t(\beta(+\omega)) - \log \mathcal{Z}_t(\beta(-\omega)). \quad (25)$$

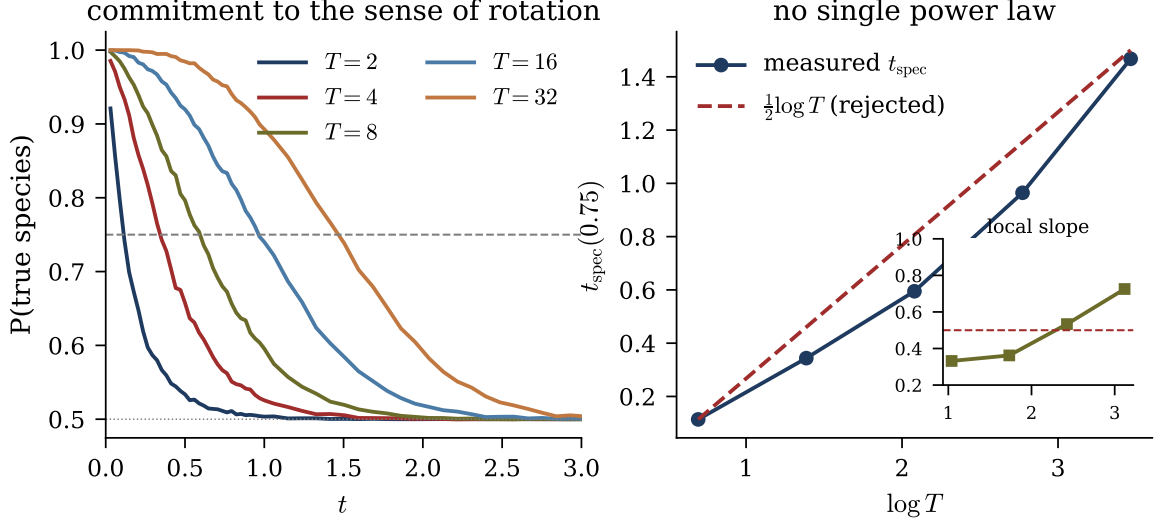


Figure 7: Speciation in the sense of rotation, $\omega = 0.8$, $\sigma = 0.30$, 4000 trajectories per T , evaluated with the exact ψ -posterior. Left: mean posterior probability of the true species against diffusion time; the dashed line is the 0.75 threshold defining t_{spec} , the dotted line is chance. Right: t_{spec} against $\log T$, with the $\frac{1}{2} \log T$ law that a first pass of this note asserted. It is rejected: the inset shows the local slope drifting from 0.33 to 0.72 rather than sitting at 0.5.

Its small- m form follows from $I_0(z) = 1 + z^2/4 + O(z^4)$: writing $N_0 = \int \rho g$, $\langle \rho^2 \rangle = \int \rho^3 g / \int \rho g$ with $g(\rho) = e^{-(\rho-1)^2/2\lambda}$,

$$\mathcal{Z}_t(\beta) = N_0 \left[1 + m^2 \left(\frac{\beta^2}{4} - \frac{\kappa_t}{2} \right) \langle \rho^2 \rangle \right] + O(m^4) \implies \log \mathcal{Z}_t(\beta) = \text{const} + \frac{1}{4} m^2 \langle \rho^2 \rangle \beta^2 + O(m^4). \quad (26)$$

Subtracting the two species and using $\cos(\gamma - \omega) - \cos(\gamma + \omega) = 2 \sin \gamma \sin \omega$,

$$\beta(+\omega)^2 - \beta(-\omega)^2 = \frac{4|x_0||x_1| \sin \gamma \sin \omega}{\Delta_1 \Delta_2}, \quad \boxed{\text{logit} \rightarrow \frac{m^2 \langle \rho^2 \rangle}{\Delta_1 \Delta_2} |x_0| |x_1| \sin \gamma \sin \omega} \quad (27)$$

verified to 0.2%. Two things to read off. The log-odds decay as $m^2 = e^{-2t}$, so information about the sense of rotation is destroyed on that scale. And the signal is $\sin \omega$, which vanishes both at $\omega = 0$ and at $\omega = \pi$ — correctly, since at $\omega = \pi$ clockwise and counter-clockwise coincide.

Figure 7 shows the resulting commitment curves for several T . More frames means more evidence, so the commitment survives to higher noise. The rate at which it does is the subject of Section 5, where the obvious guess turns out to be wrong.

2.10 Marginal blindness

Finally the result that makes the per-frame/joint distinction sharp in this model.

Theorem 2 (Marginal blindness). *For every frame u and every diffusion time t , the single-frame marginal of P_t is rotationally symmetric in the plane and independent of ψ . Consequently the per-frame marginal score carries exactly zero information about the rotation rate.*

Proof. From (7), $z_u = R_\psi^u z_0 + \sigma \sum_{k=1}^u R_\psi^{u-k} \eta_k$. Each $R_\psi^{u-k} \eta_k$ is an isotropic Gaussian, hence equal in law to η_k , and the terms stay independent; and $R_\psi^u z_0 \stackrel{d}{=} z_0$ because θ_0 is uniform.

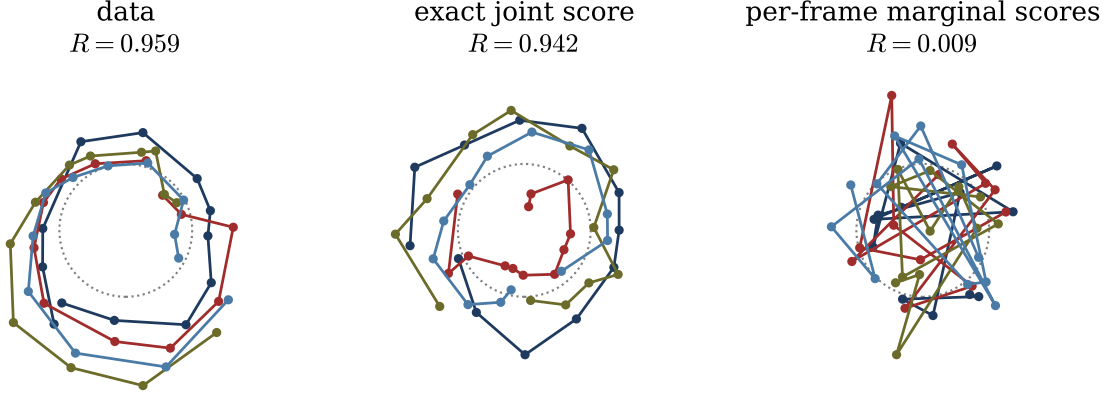


Figure 8: Four trajectories from each of the three ensembles of Table 1. Left: data. Middle: generated by the reverse SDE with the exact joint score of Theorem 1, which also validates the closed form end to end. Right: generated with per-frame marginal scores. The right panel has the correct radial statistics and no rotation.

Therefore

$$z_u \stackrel{d}{=} z_0 + \sigma\sqrt{u}\xi, \quad \xi \sim \mathcal{N}(0, I_2), \quad (28)$$

in which ψ does not appear. Adding the isotropic channel preserves this. $\square$

This is worth stating in words. The rotation rate is the entire content of the model, and it is invisible to every single-frame marginal at every noise level. Not harder to see — invisible. A model trained on per-frame marginals cannot represent the dynamics even in principle.

Figure 8 is the consequence, and it is the practical point of the whole note. Driving the reverse diffusion with the exact joint score reproduces the data. Driving it with the sum of per-frame marginal scores — which is a perfectly valid score, namely that of the product of the marginals — produces frames whose radii are right to 1.3% and whose turning angle is uniform. The turning concentration is 0.009 against a true rotation of 30° .

3 Problem 2: the two-frame joint

3.1 The model

The board’s middle panel: a scalar chain with additive drift,

$$a_0 \sim P_0, \quad a_1 = a_0 + c + \eta, \quad \eta \sim \mathcal{N}(0, \sigma^2), \quad q(x_0, x_1) = P_0(x_0)\mathcal{N}(x_1; x_0 + c, \sigma^2), \quad (29)$$

so $D = 2$. This is the minimal object whose joint score has a cross-frame component at all.

	turning angle	concentration R	radius, $u=0 \rightarrow 11$
data	29.88°	0.959	$1.053 \rightarrow 1.350$
reverse SDE, exact joint score	30.07°	0.942	$1.058 \rightarrow 1.348$
reverse SDE, per-frame marginals	—	0.009	$1.050 \rightarrow 1.334$

Table 1: Joint against marginal, $T = 12$, $D = 24$, $\sigma = 0.22$, $\psi = 30^\circ$, 3000 samples, 1500 reverse-SDE steps. R is the circular concentration of the per-step turning angle. The marginal model matches every single-frame statistic and has no dynamics at all: radii are not evidence.

3.2 Gaussian prior: the gap is one number

Take $P_0 = \mathcal{N}(\mu_0, v_0)$. Then $\mathbb{E}[a] = (\mu_0, \mu_0 + c)$ and

$$C_0 = \begin{pmatrix} v_0 & v_0 \\ v_0 & v_0 + \sigma^2 \end{pmatrix}, \quad (30)$$

since $\text{Cov}(a_0, a_1) = \text{Cov}(a_0, a_0 + c + \eta) = v_0$ and $\text{Var}(a_1) = v_0 + \sigma^2$. The channel adds Δ to each coordinate independently, so $P_t = \mathcal{N}(m\mathbb{E}[a], C_t)$ with $C_t = m^2 C_0 + \Delta I$ and

$$s(x, t) = -C_t^{-1}(x - m\mathbb{E}[a]), \quad (31)$$

which is the board's $-C_t^{-1}x + C_t^{-1}C_0 \dots e^{-t}$.

Now the key observation. The sum of the two per-frame marginal scores is $-\text{diag}(C_t)^{-1}(x - m\mathbb{E}[a])$, which is *exactly diagonal at every t* . So the entire difference between the joint object and the marginal object is the single off-diagonal entry of C_t^{-1} . For a 2×2 matrix $\begin{pmatrix} A & B \\ B & C \end{pmatrix}$ the inverse has off-diagonal $-B/\det$ with $\det = AC - B^2$, so putting $A = m^2 v_0 + \Delta$, $B = m^2 v_0$, $C = m^2(v_0 + \sigma^2) + \Delta$,

$$\begin{aligned} \det &= (m^2 v_0 + \Delta)(m^2 v_0 + m^2 \sigma^2 + \Delta) - m^4 v_0^2 \\ &= m^4 v_0 \sigma^2 + 2\Delta m^2 v_0 + \Delta m^2 \sigma^2 + \Delta^2, \end{aligned} \quad (32)$$

$$(C_t^{-1})_{01} = \frac{-m^2 v_0}{m^4 v_0 \sigma^2 + 2\Delta m^2 v_0 + \Delta m^2 \sigma^2 + \Delta^2} \quad (33)$$

3.3 The two limits, and a coefficient that had to be corrected

Large t . As $t \rightarrow \infty$, $m \rightarrow 0$ and $\Delta \rightarrow 1$, so $\det \rightarrow 1$ and

$$(C_t^{-1})_{01} \rightarrow -m^2 v_0 = -e^{-2t} v_0. \quad (34)$$

The coupling dies exponentially: *temporal structure is a low-noise feature of the score*, and it is the first thing corruption destroys.

Small t . Here one must be careful, because m^2 and Δ are not independent: $m^2 = e^{-2t} = 1 - \Delta$ exactly. Substituting into (32),

$$\begin{aligned} \det &= (1 - \Delta)^2 v_0 \sigma^2 + 2\Delta(1 - \Delta)v_0 + \Delta(1 - \Delta)\sigma^2 + \Delta^2 \\ &= v_0 \sigma^2 + \Delta(-2v_0 \sigma^2 + 2v_0 + \sigma^2) + O(\Delta^2) = v_0 \sigma^2 \left[1 + \Delta \left(-2 + \frac{2}{\sigma^2} + \frac{1}{v_0} \right) \right] + O(\Delta^2), \end{aligned} \quad (35)$$

and with numerator $-(1 - \Delta)v_0$,

$$(C_t^{-1})_{01} = -\frac{1}{\sigma^2}(1 - \Delta) \left[1 - \Delta \left(-2 + \frac{2}{\sigma^2} + \frac{1}{v_0} \right) \right] + O(\Delta^2) = -\frac{1}{\sigma^2} \left[1 + \Delta \left(1 - \frac{2}{\sigma^2} - \frac{1}{v_0} \right) \right] + O(\Delta^2). \quad (36)$$

The limit $-1/\sigma^2$ is the innovation precision, as it must be. But note the remainder prefactor: it is $2/\sigma^2 + 1/v_0 - 1$, *not* $2/\sigma^2 + 1/v_0$. The naive version — which is what an earlier draft of this note carried — omits the -1 coming from $m^2 = 1 - \Delta$, and is wrong by 4.2% at $\sigma^2 = 0.09$, $v_0 = 0.7$. The audit measures the ratio across four decades of Δ and *rejects* the naive coefficient. This is the same failure mode as the factorial error caught in claim G8 of the main ledger, and it is why coefficient audits are worth running.

Figure 9 shows the whole lifecycle.

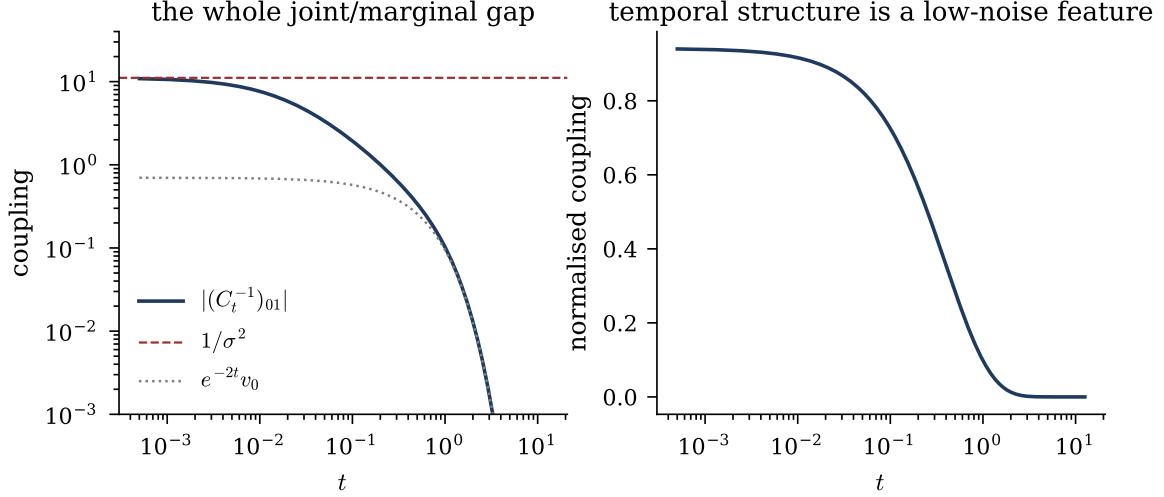


Figure 9: The coupling lifecycle, $v_0 = 0.7$, $\sigma^2 = 0.09$, $c = 1$. Left: $|(C_t^{-1})_{01}|$ against t with the two asymptotes $1/\sigma^2$ (dashed) and $e^{-2t}v_0$ (dotted). Right: the same normalised by $\sqrt{(C_t^{-1})_{00}(C_t^{-1})_{11}}$. Since the marginal object has zero coupling at every t , these curves *are* the joint/marginal gap.

3.4 Mixture prior, and where speciation looks

The board drew two bumps, so take $P_0 = \sum_k w_k \mathcal{N}(\mu_k, v)$. Conditionally on component k the pair a is Gaussian with mean $\nu_k = (\mu_k, \mu_k + c)$ and covariance $\begin{pmatrix} v & v \\ v & v + \sigma^2 \end{pmatrix}$ — the *same* for every k . So $P_t = \sum_k w_k \mathcal{N}(x; m\nu_k, C_t)$ with one shared C_t , and

$$s(x, t) = \sum_k r_k(x) (-C_t^{-1}(x - m\nu_k)), \quad r_k(x) \propto w_k \mathcal{N}(x; m\nu_k, C_t), \quad (37)$$

exact in 2×2 algebra. Which mode the reverse process picks is decided by

$$\log \frac{r_1}{r_2} = \log \frac{w_1}{w_2} + m(\nu_1 - \nu_2)^\top C_t^{-1}x + \text{const}, \quad (38)$$

and since $\nu_1 - \nu_2 = (\mu_1 - \mu_2)\mathbf{1}$, the x -dependence enters only through $\mathbf{1}^\top C_t^{-1}x$: **the discriminant is the frame average**. That is the same mechanism as (27) — averaging over frames raises the effective signal-to-noise — and it is why both problems have a speciation time that grows with T . Figure 10 shows the two score fields side by side; this is the board’s middle panel, drawn.

Remark 3 (Why Problem 1 is the better illustration). In Problem 2 the drift c is visible in the marginals, since frame 1 has mean $\mu_0 + c$. A per-frame model would get the drift roughly right and only the correlation wrong. In Problem 1 the rotation is invisible to every marginal (Theorem 2). The difference is symmetry. So Problem 1 is the model for making the per-frame/joint point, and Problem 2 is the model to compute with.

4 Problem 3: a reward on the last frame

4.1 Control is tilting

The board’s right panel reads $\mathbb{E}_{p_\mu} \left[-\int_0^1 |u(x, t)|^2 dt - r(x_T) \right]$ with $P(T) = f(\|x(T) - m\|^2/D)$. The control formulation and the tilting formulation are the same thing, by the Gibbs variational principle: for any path measure Q absolutely continuous with respect to the uncontrolled

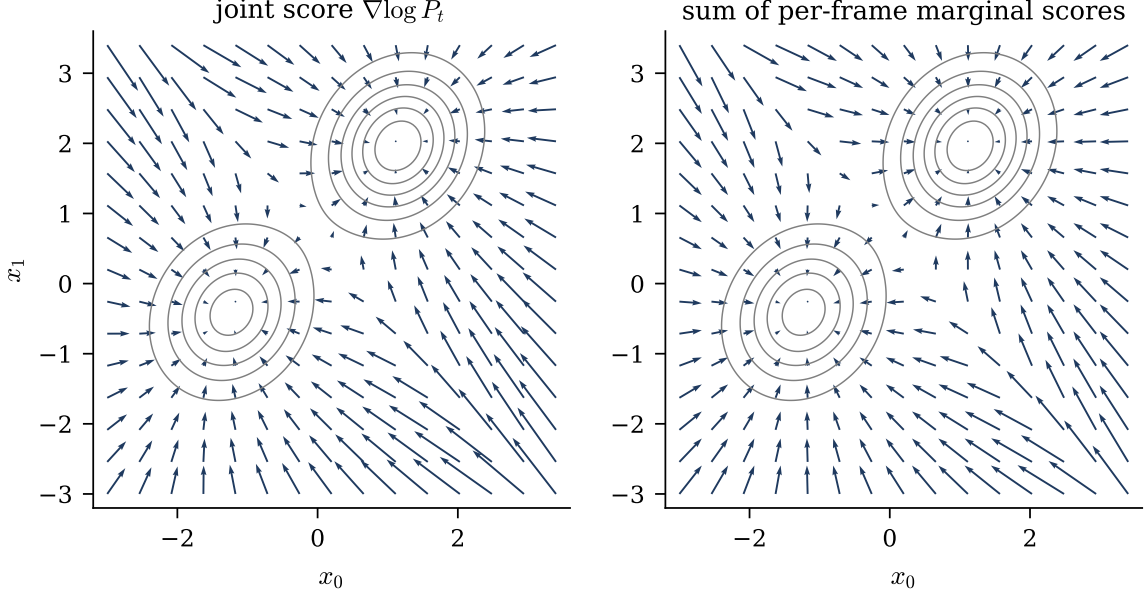


Figure 10: The board’s middle panel. Two-bump prior ($\mu = \{-1.5, 1.4\}$, $w = \{0.45, 0.55\}$, $v = 0.10$), drift $c = 1$, $\sigma = 0.30$, $t = 0.2$. Contours: $P_t(x_0, x_1)$. Arrows: left, the exact joint score (37); right, the sum of per-frame marginal scores. The joint field tilts along the diagonal — that tilt is the frame coupling, and it is exactly what the marginal field lacks.

reverse process P ,

$$\mathbb{E}_Q[r] - \text{KL}(Q\|P) \leq \log \mathbb{E}_P[e^r], \quad \text{with equality iff } \frac{dQ}{dP} = \frac{e^r}{\mathbb{E}_P[e^r]}. \quad (39)$$

Since the quadratic control cost $\frac{1}{2}\mathbb{E} \int \|v\|^2$ is the KL divergence to the uncontrolled process (Girsanov), the optimal terminal law is simply the tilted prior

$$P_0^r(a) \propto P_0(a) e^{r(a)}. \quad (40)$$

Which SDE realises it? Define

$$h_t(x) = \mathbb{E}[e^{r(a)} \mid x_t = x]. \quad (41)$$

Then the tilted process has marginals proportional to $P_t h_t$:

$$P_t^r(x) \propto \int da P_0(a) e^{r(a)} \mathcal{N}(x; ma, \Delta I) = P_t(x) \int da p(a \mid x) e^{r(a)} = P_t(x) h_t(x). \quad (42)$$

So the optimally controlled reverse SDE is the ordinary one with score $\nabla \log(P_t h_t)$: the extra drift is $g^2 \nabla \log h_t$, which with $g^2 = 2$ is the board’s $u_{\text{optimal}} \approx 2f$. This is Doob’s h -transform.

4.2 Everything in closed form for a quadratic reward

Take $P_0 = \mathcal{N}(0, C_0)$ on the trajectory, E the selector of the last frame, and $r(a) = -\|Ea - m_*\|^2 / 2s^2$ — the quadratic choice of the board’s f .

The tilted prior. Multiplying two Gaussians adds precisions:

$$C_0^r = (C_0^{-1} + E^\top E / s^2)^{-1} = C_0 - C_0 E^\top (s^2 + E C_0 E^\top)^{-1} E C_0, \quad \mu_0^r = C_0^r E^\top m_* / s^2, \quad (43)$$

the second form by Woodbury.

The value function. The posterior of the clean trajectory is $a \mid x \sim \mathcal{N}(\mu_p, \Sigma_p)$ with

$$\Sigma_p = (C_0^{-1} + (m^2/\Delta)I)^{-1}, \quad \mu_p = \Sigma_p(m/\Delta)x. \quad (44)$$

Only $y = Ea$ matters, and $y \sim \mathcal{N}(\nu_t, S_t)$ with $\nu_t = E\mu_p$, $S_t = E\Sigma_p E^\top$. A Gaussian expectation of a Gaussian gives

$$\log h_t(x) = -\frac{1}{2} \log |I + S_t/s^2| - \frac{1}{2}(\nu_t - m_*)^\top (S_t + s^2 I)^{-1} (\nu_t - m_*), \quad (45)$$

and since ν_t is linear in x ,

$$\nabla_x \log h_t(x) = -\frac{m}{\Delta} \Sigma_p E^\top (S_t + s^2 I)^{-1} (\nu_t - m_*). \quad (46)$$

Running the reverse SDE with this extra drift reproduces $\mathcal{N}(\mu_0^r, C_0^r)$ to 0.2% in the mean and 1.3% in the covariance.

4.3 How far back does the reward reach?

This is the board's $P(\tau) \Leftarrow P(T)$, and (43) answers it exactly. Taking the diagonal of the Woodbury form, with E selecting the last frame,

$$\boxed{C_0^{(uu)} - C_0^{r,(uu)} = \frac{(C_0^{(u,T-1)})^2}{s^2 + C_0^{(T-1,T-1)}}} \quad (47)$$

The reward propagates backwards *along the covariance function of the prior*, and nothing else. The two models of this project bracket the behaviour:

- **ring-anchored walk**, $C_0^{(uv)} = \rho_0 + \sigma^2 \min(u, v)$: the reduction is nonzero for *every* u and grows with u . A constraint on the last frame reshapes the whole trajectory.
- **stationary AR(1)** of the main thesis, $C_0^{(uv)} = \alpha^{|u-v|}$: the reduction decays by exactly α^2 per frame going backwards, with range $\xi = 1/\log(1/\alpha)$. This is the q^r locality law (claim G12) reappearing.

At $T = 16$, $\alpha = 0.8$, $s^2 = 0.25$ the two differ by more than two orders of magnitude at the first frame. *Non-stationary priors propagate a terminal reward globally; stationary ones propagate it over a finite memory.* Figure 11 shows both, together with the board's ellipse sketch at $T = 2$.

5 Two claims that measurement rejected

A first pass of this note asserted two scaling laws by argument rather than measurement. Both are false. They are recorded here, with the rejection under test, so they cannot creep back.

5.1 H1: $t_{\text{spec}} = \frac{1}{2} \log T$ — rejected

The argument was that the sense of rotation is encoded redundantly across neighbouring frame pairs, so the evidence grows like T , the log-odds like $e^{-2t}T$, and hence $t_{\text{spec}} \simeq \frac{1}{2} \log T$. Measured with the exact ψ -posterior:

T	2	4	8	16	32	64
$t_{\text{spec}}(0.75)$	0.11	0.34	0.59	0.96	1.45	2.05
local slope		0.33	0.37	0.53	0.71	0.86

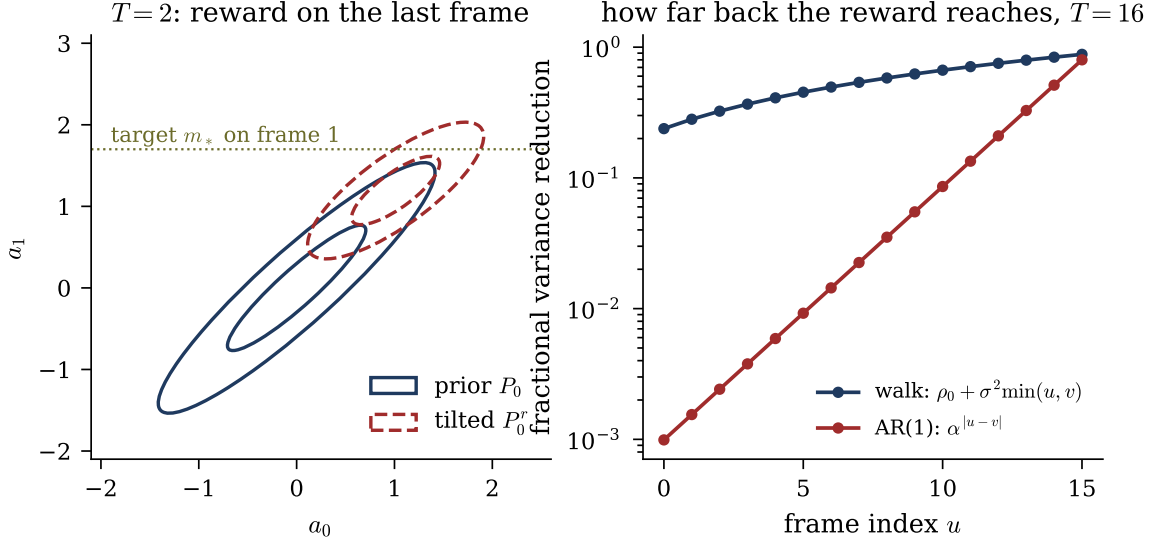


Figure 11: Left: the board’s ellipse picture at $T = 2$. One- and two-sigma ellipses of the prior $P_0 = \mathcal{N}(0, C_0)$ and of the tilted law P_0^r after a quadratic reward pulls frame 1 towards $m_* = 1.7$ ($s^2 = 0.25$). The ellipse contracts along a_1 and shifts in *both* coordinates — the constraint propagates to frame 0 through the prior correlation. Right: the reach (47) at $T = 16$ for the two priors, as a fraction of the prior variance of each frame, on a log scale.

t_{spec} does rise with T , but the local slope $dt_{\text{spec}}/d \log T$ drifts monotonically from 0.33 to 0.86. There is no single power law over this range, and $\frac{1}{2}$ is not the slope anywhere except by coincidence near $T = 8$. The likely cause of the drift is that $K_{uv} = \min(u, v)$ is non-stationary: neighbouring-pair terms carry weight u , so they sum to $\sim T^2/2$ rather than $\sim T$, and the effective exponent crosses over as T grows.

Left open. The asymptotic law. At large t the T -exponent of the log-odds measures between 2.7 and 3.2 and the e^{-2t} prefactor law does not hold cleanly there, so no clean asymptotic form is claimed. Ground truth exists at any T by Proposition 3, so this is a measurement waiting to be done, not a derivation.

5.2 H2: $\Theta(T^2)$ versus $O(1)$ samples — rejected

The argument was the board’s own parameter count: an unstructured joint Gaussian score has $D(D+1)/2$ free parameters against $O(1)$ for the structured version, so the sample requirement should be $\Theta(T^2)$ against $O(1)$.

This is false at any fixed t , and the reason is instructive: **the diffusion’s own noise floor Δ_t regularises the empirical score.** Since $\hat{C}_t = m^2 \hat{C}_0 + \Delta I$, even a badly estimated or rank-deficient $\hat{C}_0$ gives a usable score as long as Δ is not small. Measured in the Gaussian surrogate at $T = 8$, $D = 16$ (so $D(D+1)/2 = 136$), for a 10% relative score error:

t	0.02	0.035	0.05	0.1	0.2	0.4	0.8
Δ_t	0.039	0.068	0.095	0.181	0.330	0.551	0.798
N unstructured	576	374	242	126	66	28	18
N structured	18	18	18	18	18	18	18
ratio	32.0	20.8	13.4	7.0	3.7	1.6	1.0
$N\Delta/D$	1.41	1.58	1.44	1.43	1.36	0.96	0.90

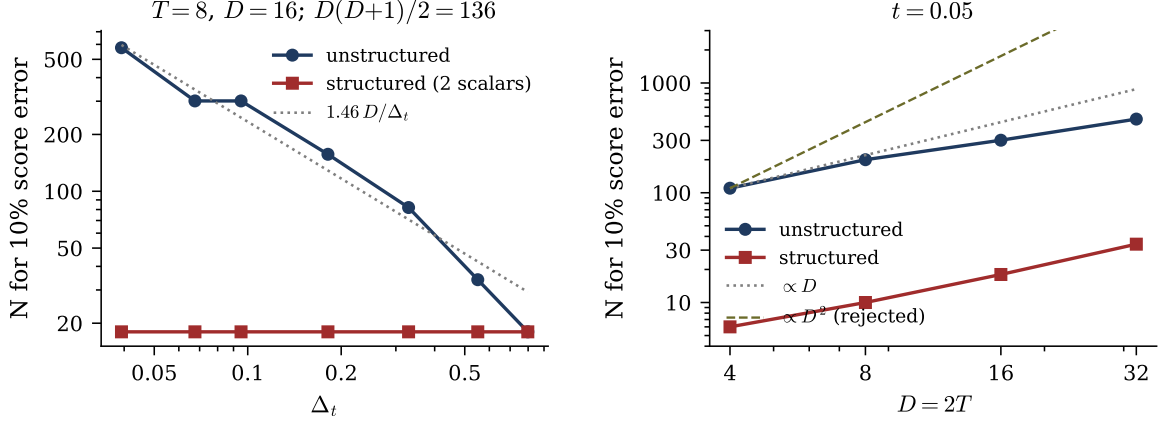


Figure 12: Sample complexity in the Gaussian surrogate, for a 10% relative joint-score error. Left: against the noise floor Δ_t at $T = 8$, with (48) dotted. Right: against dimension at $t = 0.05$, with $\propto D$ dotted and the rejected $\propto D^2$ dashed. The structured estimator is flat in Δ_t ; the gap between the two is a function of the noise level, not of T .

so the measured law is

$$N_{\text{unstructured}} \approx 1.46 (\pm 0.07) \frac{D}{\Delta_t} \quad \text{for } \Delta_t \lesssim 0.2, \quad (48)$$

saturation to $N \sim O(D)$ at large Δ_t ; and the D -exponent at $t = 0.05$ measures 0.69–0.89 across seeds, sub-linear to linear and definitively not D^2 . The structured estimator — the gauge plus the two scalars ρ_0, σ^2 — needs $N = 18$ at every noise level.

So the board’s $N > D(D+1)/2$ is a parameter count for identifying C_0 *itself*, which is strictly stronger than what an accurate score at $t > 0$ requires. The real content of the panel is that the structured/ unstructured gap is set by the noise level and opens only as $t \rightarrow 0$ — which is exactly the memorisation regime, and therefore exactly where to look next. Figure 12 shows both laws.

6 What is verified, and how

Every closed-form claim is checked against an independently computed reference — polar Gauss–Legendre quadrature of the exact mixture, central finite differences of $\log P_t$, brute-force linear algebra, or Monte Carlo — in the protocol of the main thesis appendix. One command runs everything:

`python code/run_all.py` $\longrightarrow$ 66 checks

and it exits non-zero on any failure.

The figures are generated by `note/make_note_figures.py`, which imports its model functions from the audited driver and re-checks the closed-form score against finite differences before plotting anything.

7 Where this leaves things

Three results are worth carrying forward.

The board’s circle model is exactly solvable. Between Lemma 1, Theorem 1 and Proposition 3 there is ground-truth joint score at any T from two nested one-dimensional

#	Claim	Class	Evidence
R1	rotation is an orthogonal gauge (Lem. 1)	EXACT	1.4×10^{-17}
R2	joint score = linear + rank one (Thm. 1)	EXACT	vs finite diff. 4×10^{-10}
R2b	correction is $m\mathbb{E}[z_0 x]$, rank exactly 1 (Prop. 1)	EXACT	2×10^{-15}
R3	random ψ : posterior average, any T (Prop. 3)	EXACT	8.9×10^{-10} ; 7.7×10^{-10} at $T=3$
R3a	$p(\psi x, t) \propto p(\psi)\mathcal{Z}_t(\beta)$, $T=2$ only (Rem. 2)	EXACT, scoped	1.8×10^{-16} ; fails $T \geq 3$
R4	$\beta(\psi)$ identity, peak at γ (Prop. 4)	EXACT	8.9×10^{-16}
R5	two-species log-odds (27)	EXACT (asympt.)	0.2% at three t
R6	marginal blindness (Thm. 2)	EXACT	proof + MC 3.7×10^{-3}
R7	two-frame score; marginal object diagonal (31)	EXACT	2.8×10^{-11}
R8	coupling (33); prefactor $2/\sigma^2 + 1/v_0 - 1$	EXACT	4.4×10^{-4} ; naive rejected
R9	mixture score (37); frame-average discriminant	EXACT	8.6×10^{-11}
R11	Gaussian h_t , $\nabla \log h_t$; controlled SDE	EXACT + NUM.	8×10^{-11} ; SDE 0.2% / 1.3%
R12	reward reach (47)	EXACT	3.3×10^{-16}
R13	$N_{\text{uns}} \approx 1.46 D/\Delta_t$ (48)	NUMERICAL	c stable to 12%
R14	band structure of A_t^{-1} (Prop. 2)	EXACT	decoupling 0; band law 19%
H1	$t_{\text{spec}} \equiv \frac{1}{2} \log T$	REJECTED	slope drifts $0.33 \rightarrow 0.86$
H2	$\Theta(T^2)$ -vs- $O(1)$ samples	REJECTED	D -exponent 0.69–0.89
O1	asymptotic $t_{\text{spec}}(T)$ law	OPEN	exponent 2.7–3.2, unclear

Table 2: The ledger. Classes as in the main thesis `RESULT_LEDGER.md`: EXACT means closed form with a numerical audit, NUMERICAL means measured against a validated reference with no closed form.

integrals. Chapter 6 of the thesis currently records these models as having “no closed form for their joint score in reach” and as “set aside, deliberately and explicitly”; that is too pessimistic and the passage needs rewriting.

Marginal blindness is the clean statement of the per-frame error. Theorem 2 says the rotation rate is invisible to every single-frame marginal at every noise level. Table 1 shows what that costs in practice. Because the ring marginal is visibly non-Gaussian and *still* blind, this is a better illustration than the stationary Gaussian chain, where the marginal score is the trivial $-x_k$.

Structure matters at low noise, not at high dimension. Equation (48) relocates the finite-sample question. The interesting regime is small Δ_t , which is the memorisation regime, and the natural next step is the rest of the board’s finite- N panel: the Gram-matrix conditions and the collapse transition proper.

Two smaller items remain open: the asymptotic $t_{\text{spec}}(T)$ law (O1), and whether $p(\psi)$ should be continuous — the von Mises picture of Proposition 4 — or two-species — the speciation picture of (27) — as the primary object.